

Aldol condensation

Mannich Reaction

Cannizaro reaction

Fisher's esterification

Steglich esterification

**Activation of carboxylic
acids**

Peptide conjugation

Staudinger's reaction

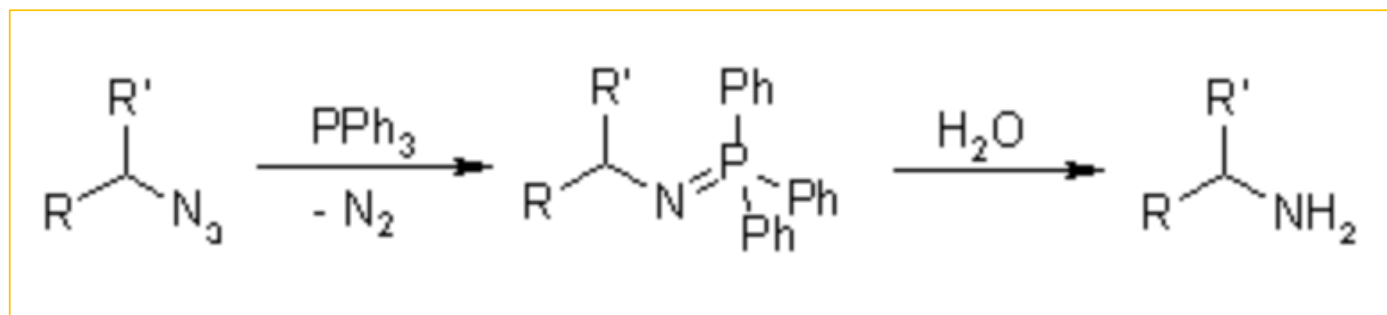
Click reaction

Wittig reaction

Sonogashira coupling

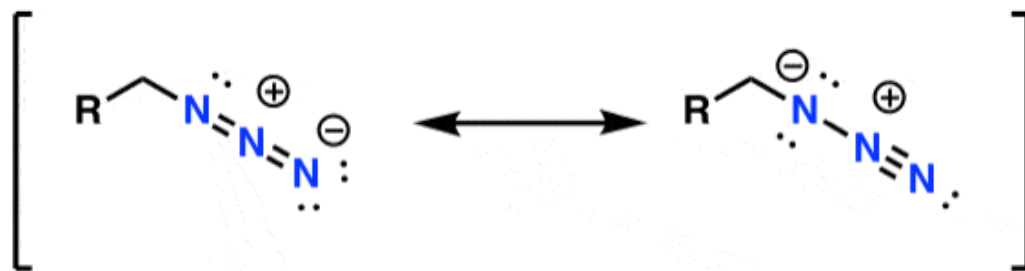
Suzuki Heck coupling

Staudinger's reaction



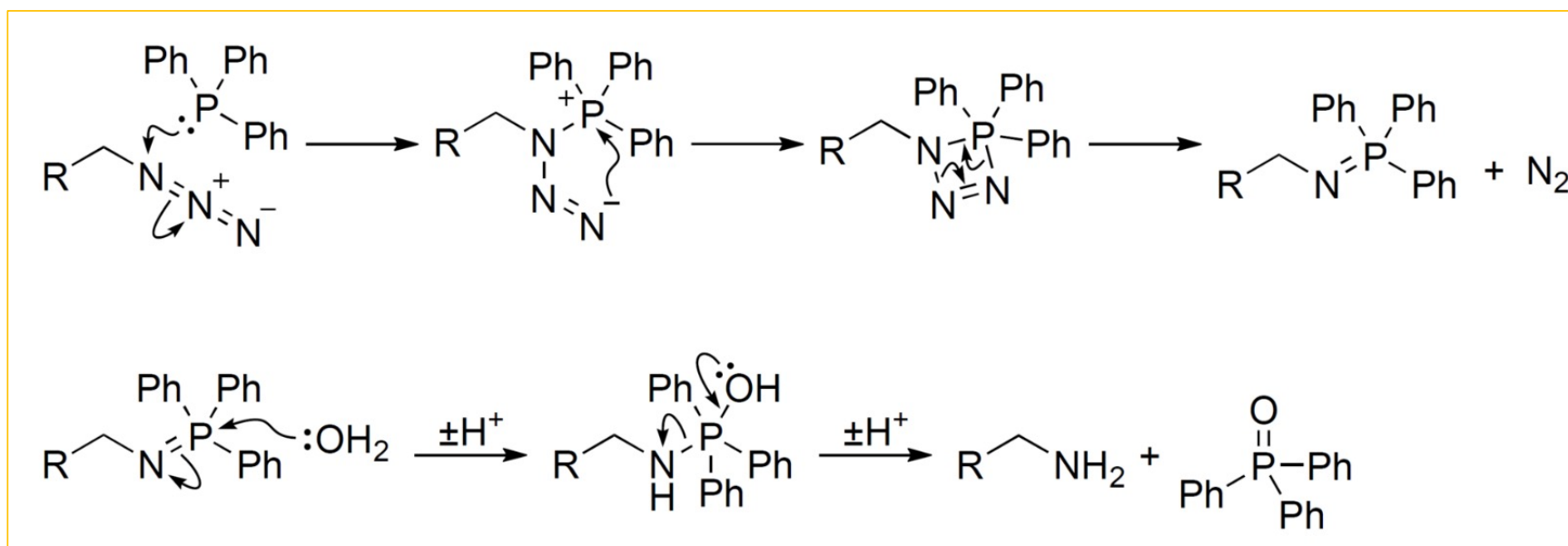
Azido $\rightarrow$ Amine
via iminophosphorane

Resonance forms of organic azides



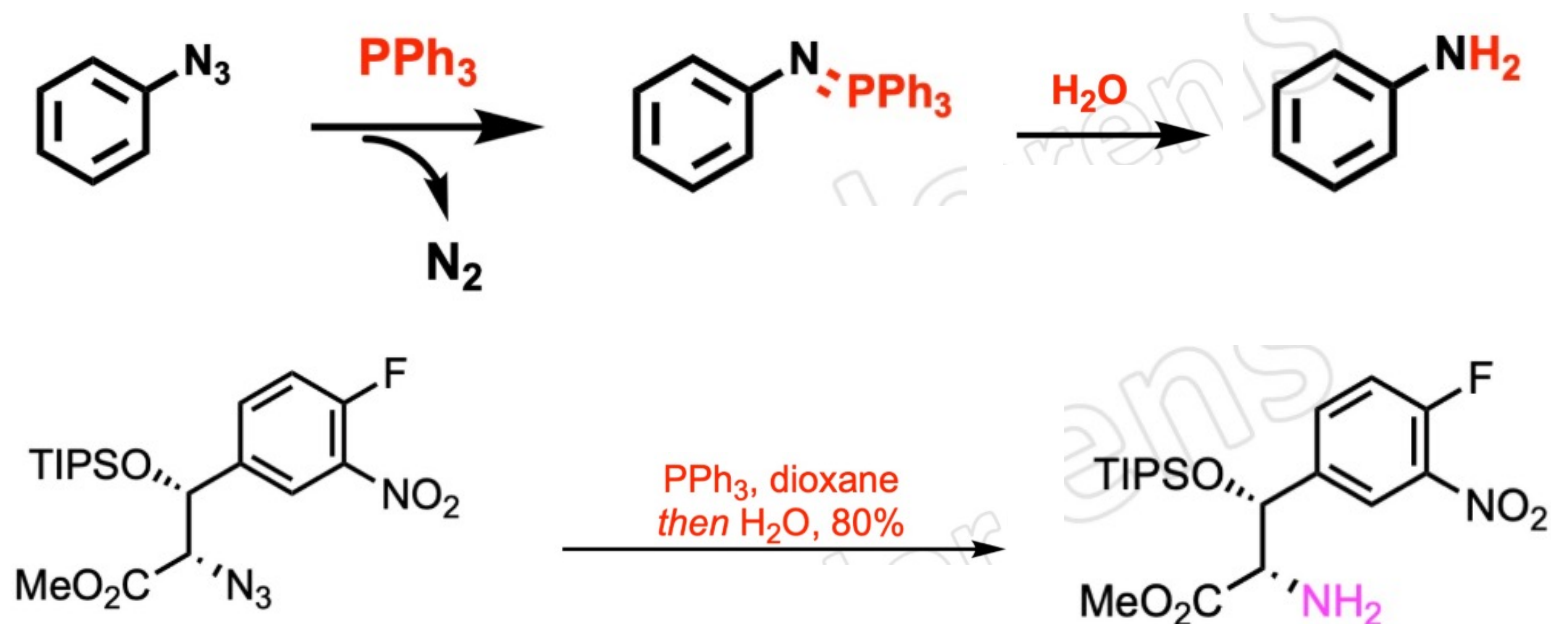
Mechanism

iminophosphorane

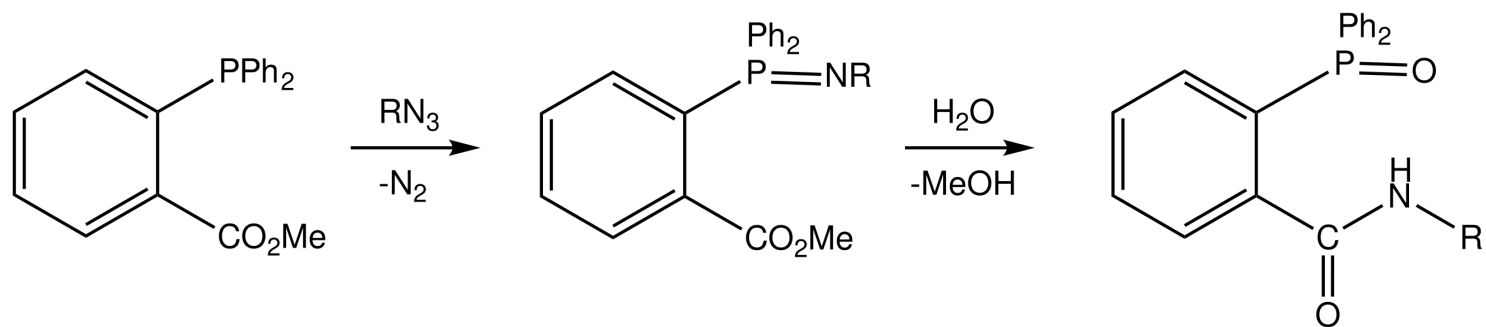


Triphenyl phosphonium oxide

Examples of Staudinger's reaction



The basic foundation for DNA sequencing (Sequencing by synthesis)

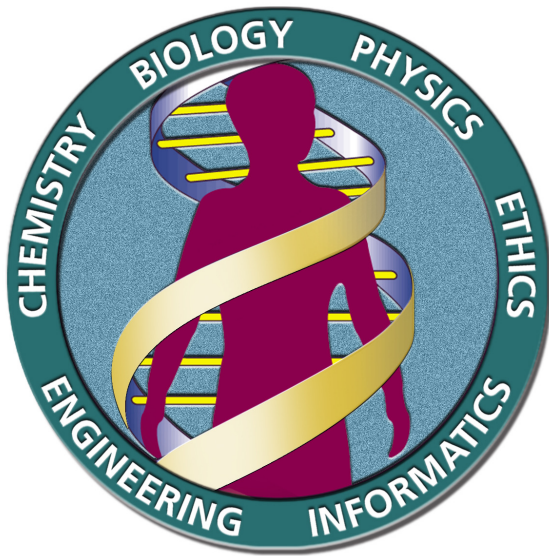


Ester (ortho) and phosphine

Staudinger's ligation

Very important in chemical biology

Human Genome Project



First complete sequence
of human genome (2022)

<https://www.nature.com/articles/d41586-025-02102-y>

[nature](#) > [news](#) > [article](#)

NEWS | 02 July 2025

First human genome from ancient Egypt sequenced from 4,800-year-old teeth

Forty years after the first effort to extract mummy DNA, researchers have finally generated a full genome sequence from an ancient Egyptian.



The Ancient Egyptian Old Kingdom (2686–2125 bc) produced many lasting artefacts — but little DNA has survived. Credit: Azoor Photo/Alamy



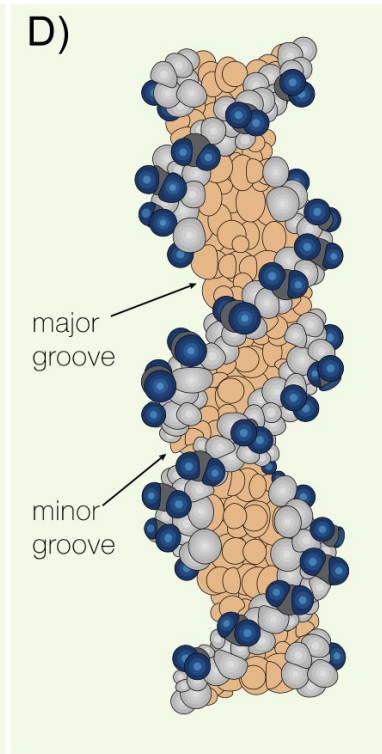
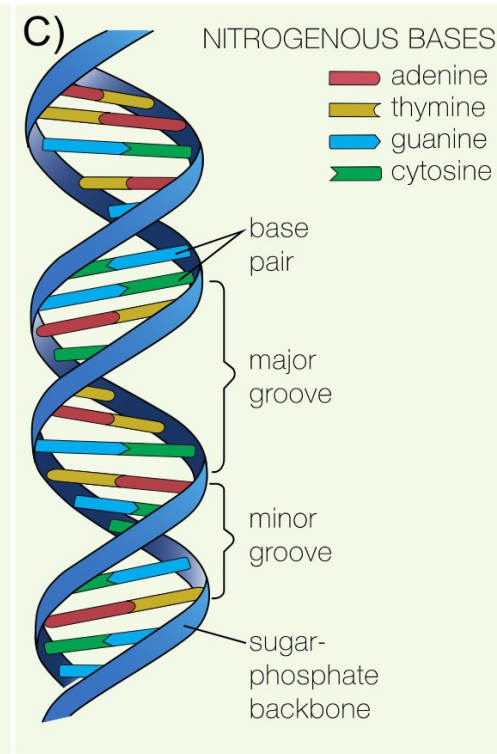
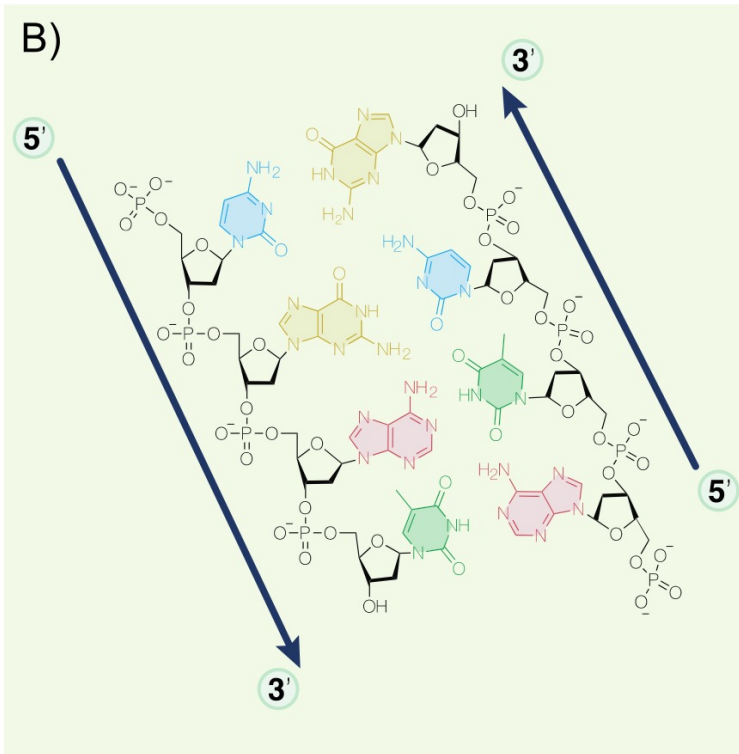
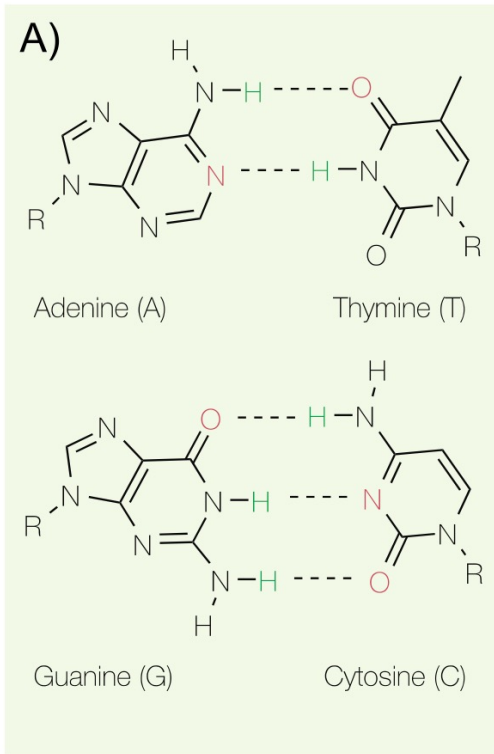
Har Gobind Khorana
Nobel Prize in Medicine (1968)

- First to chemically synthesize oligonucleotides
- First to complete total synthesis of functional gene outside living organisms



Shankar Balasubramaniam
Nobel Prize?

Scientific founder of Solexa
And Biomodal



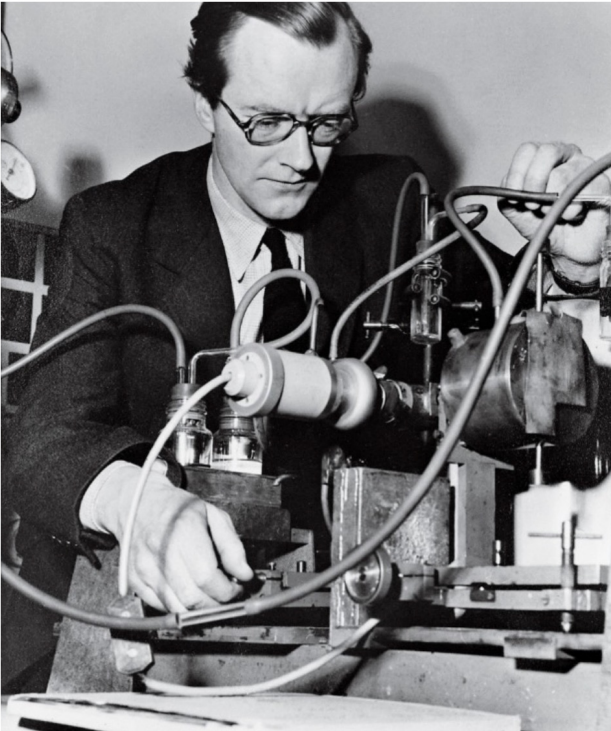
A -- T

G --- C

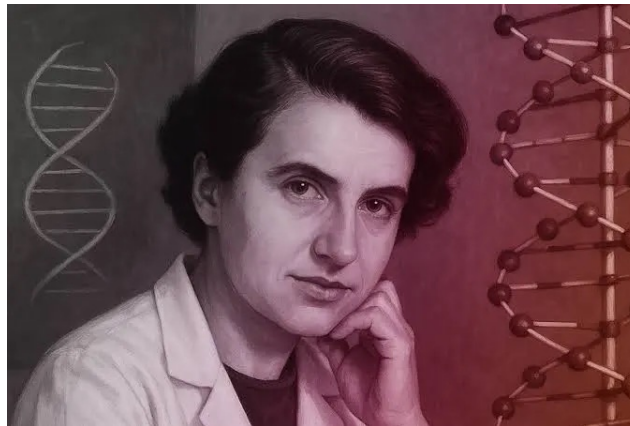
Watson-Crick-Francis

Human DNA

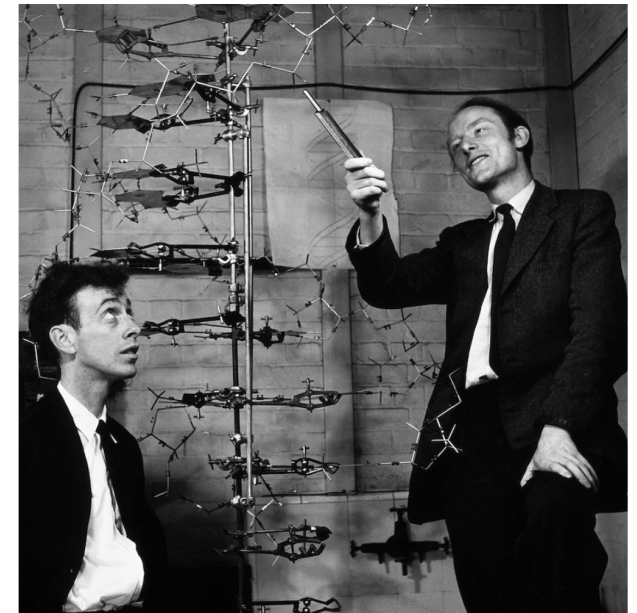
Nobel Prize in 1962: Double-helix structure of DNA



Maurice Wilkins with X-ray crystallographic equipment, ca. 1954.
King's College London and Horace Freeland Judson



The Dark Lady of DNA



James Watson and Francis Crick with their DNA model at the Cavendish Laboratory in 1953. To request permission to use this photo, please visit the [Science Photo Library website](https://www.sciencephoto.com).



DNA melting temperature: Double helix breaks into single strand DNAs

DNA Replication

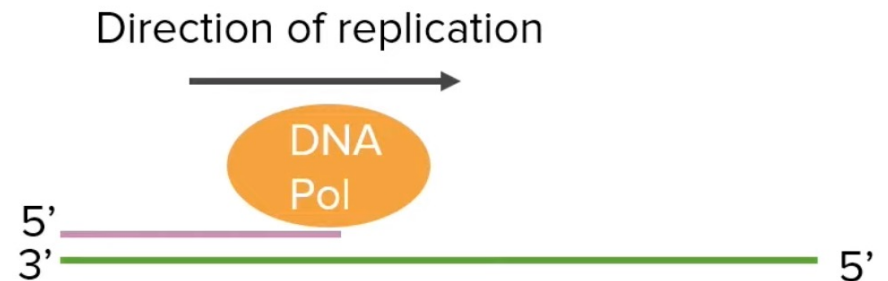
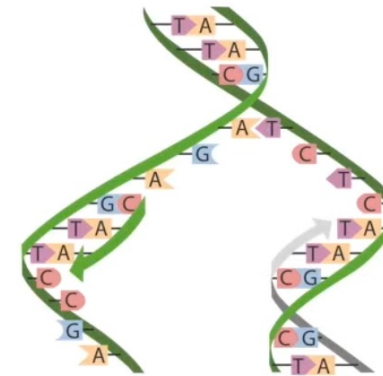
DNA Polymerase

5' to 3' Polymerase

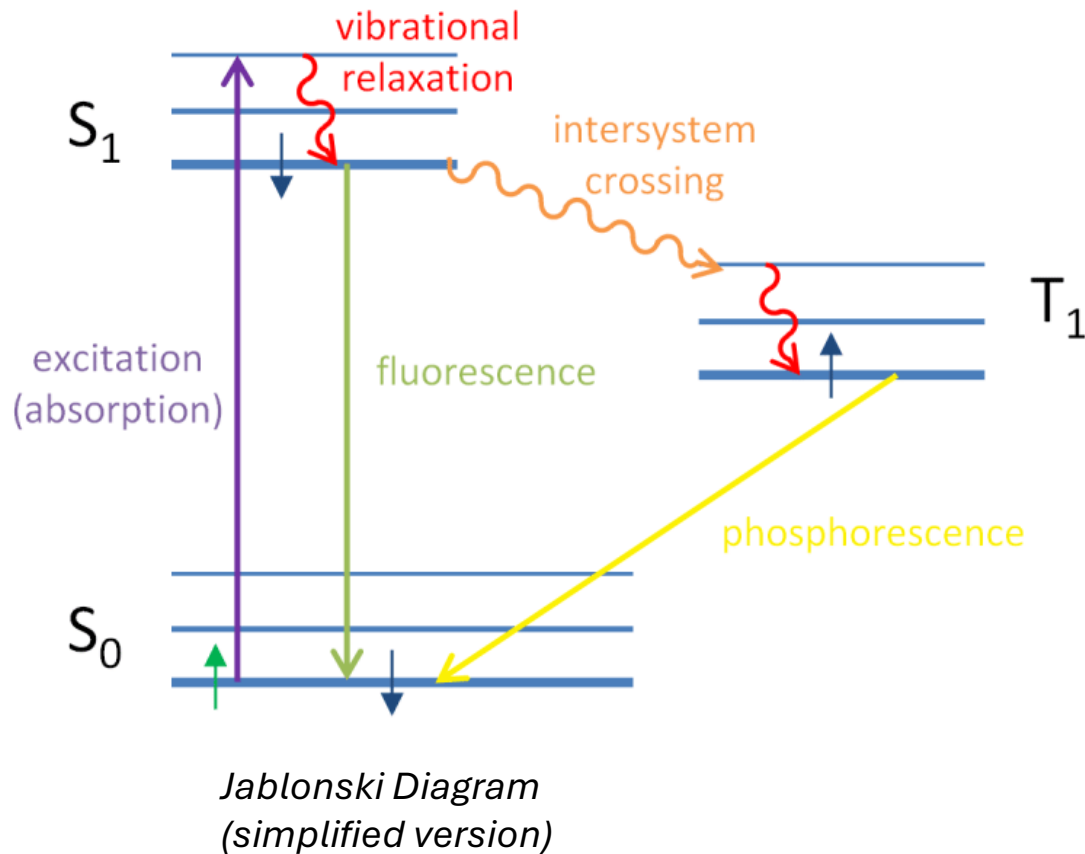
- Read base in opposite strand
- Select proper bases to put on new strand
- Join new base to new strand with phosphodiester bond

3' to 5' Exonuclease (Proofreading)

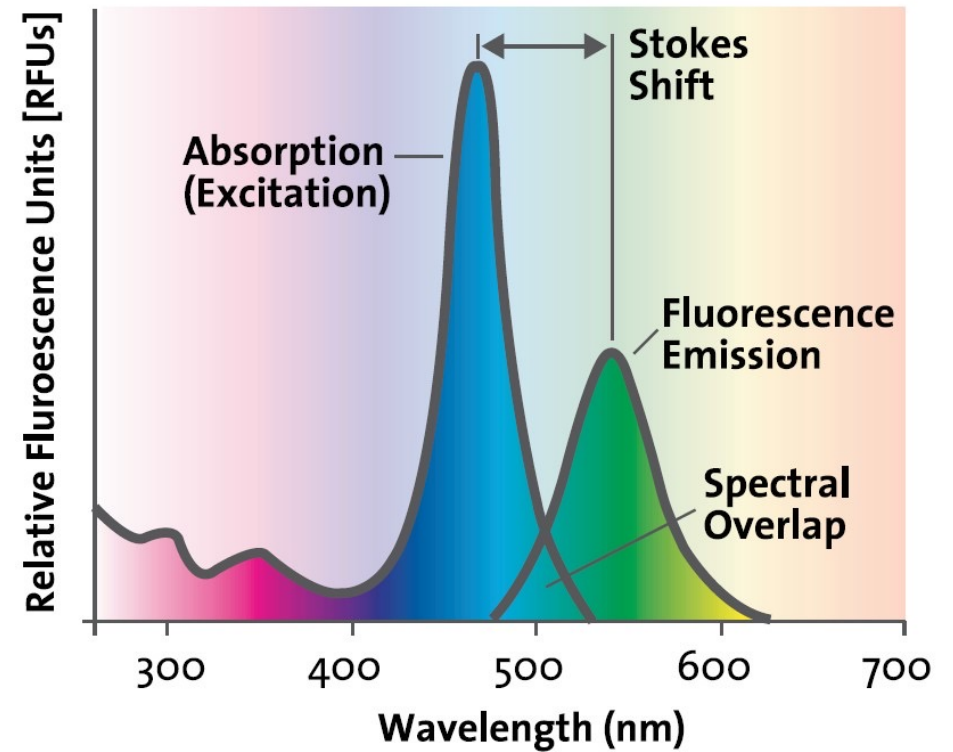
- “Reads” base pair just created for correctness
- Removes if mismatched
- Improves fidelity of copying DNA 100-fold



Fluorescence



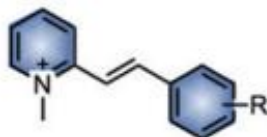
Excitation and Emission Spectral Profiles



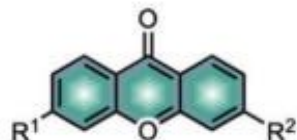
Quinoline



Styrene



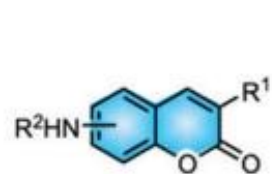
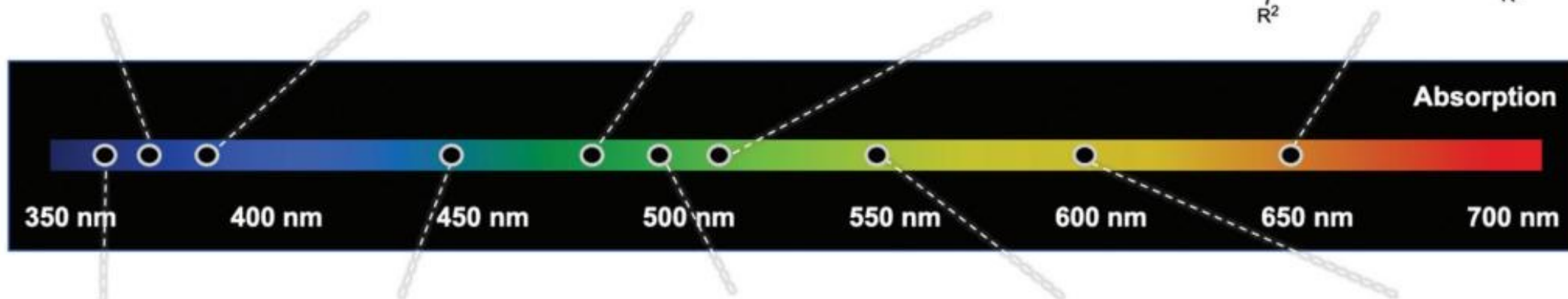
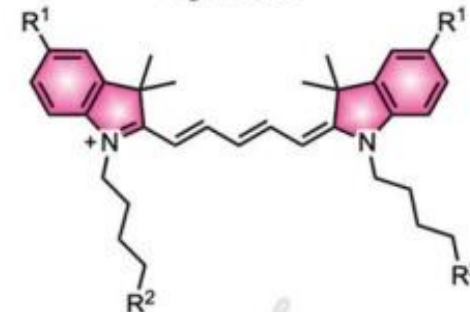
Xanthone



BODIPY



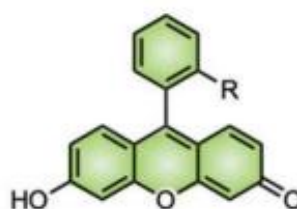
Cyanine



Coumarin



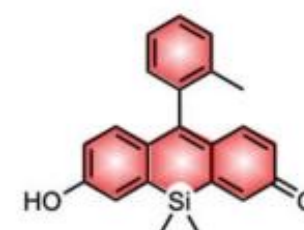
Naphthalimide



Fluorescein



Rhodamine



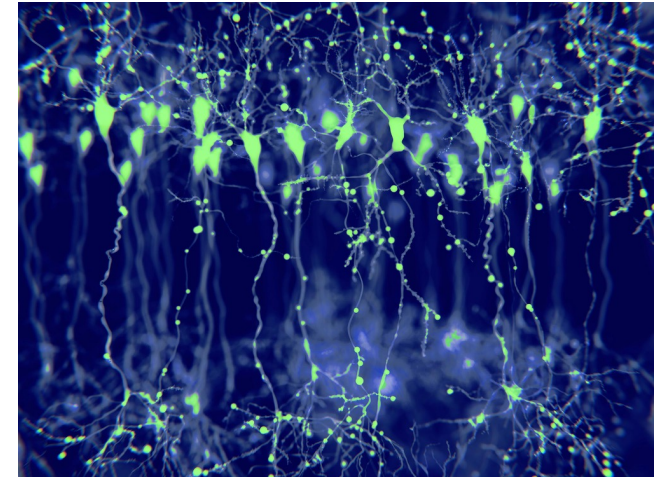
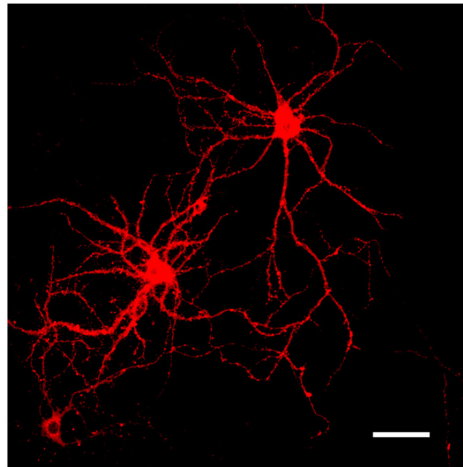
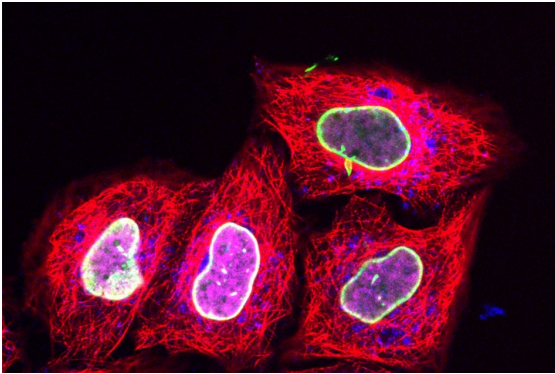
TokyoMagenta



Shine with UV or visible light

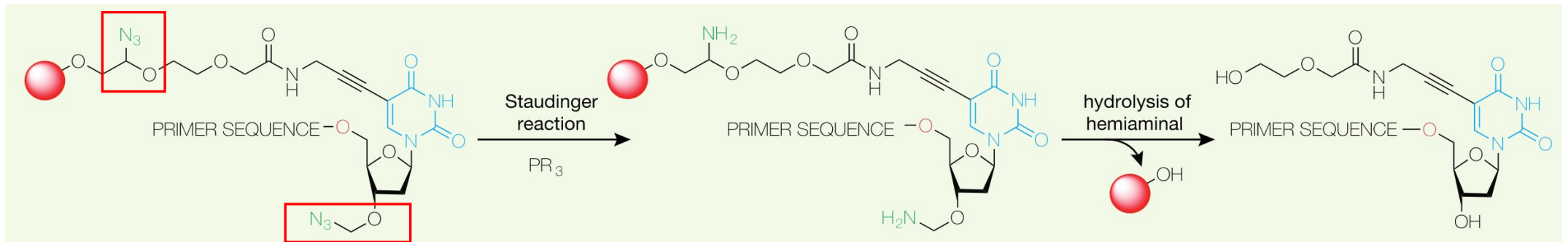
Live Cell Imaging – Confocal and fluorescence microscopy

Visualize cells and components using fluorophores

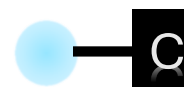
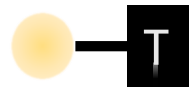
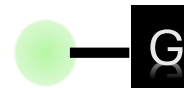
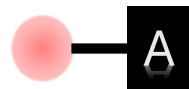


<https://www.youtube.com/watch?v=XDcf6P9Fpww>

Tagging of **biomolecules** with fluorophores

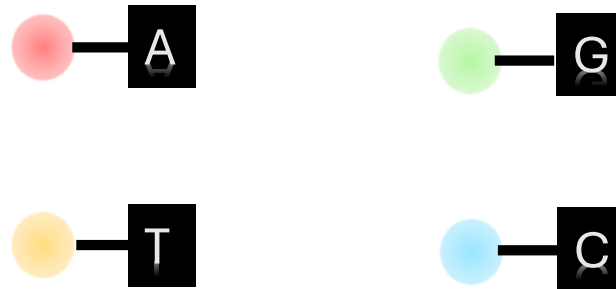


 Fluorophore

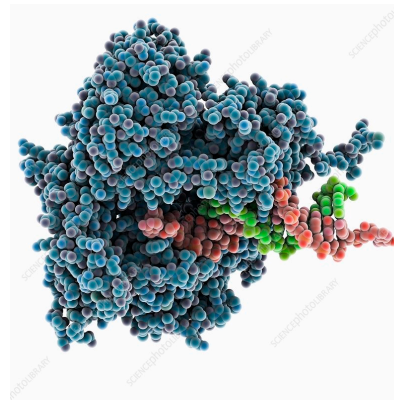




Nucleotides



DNA polymerase



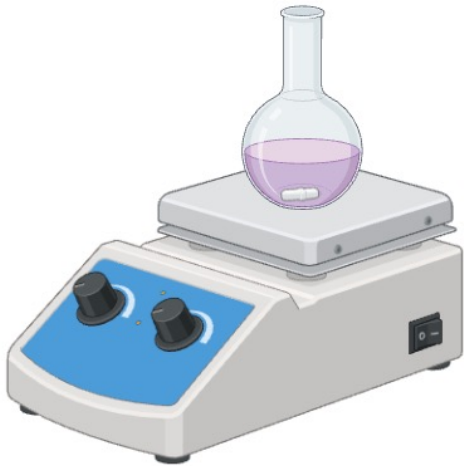
●	A	T
●	G	C
●	A	T
●	T	A
●	C	G
●	C	G
●	G	C
●	T	A
●	A	T

DNA sequencing method: Sequence by synthesis

<https://www.youtube.com/watch?v=fCd6B5HRaZ8>

<https://www.youtube.com/watch?v=fCd6B5HRaZ8>

Can we use chemical reactions in living cells?

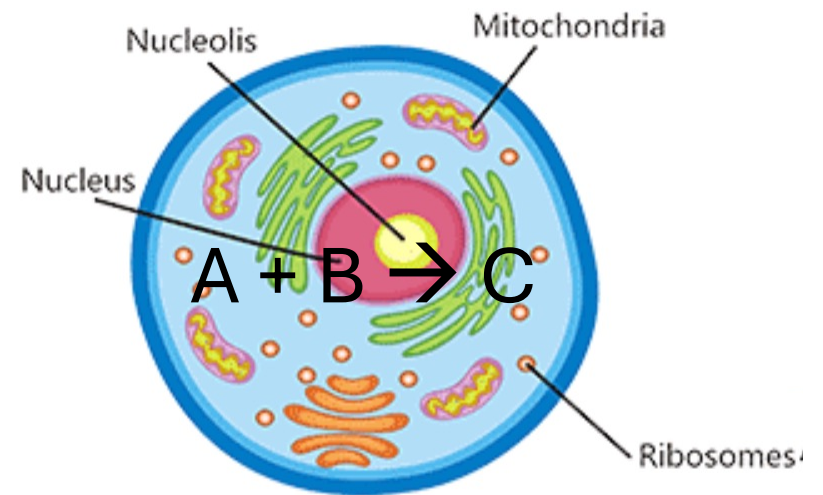
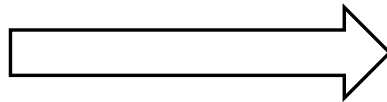


Temperature

pH

Only A and B

NO ADDITIONAL MOLECULES



Eukaryote

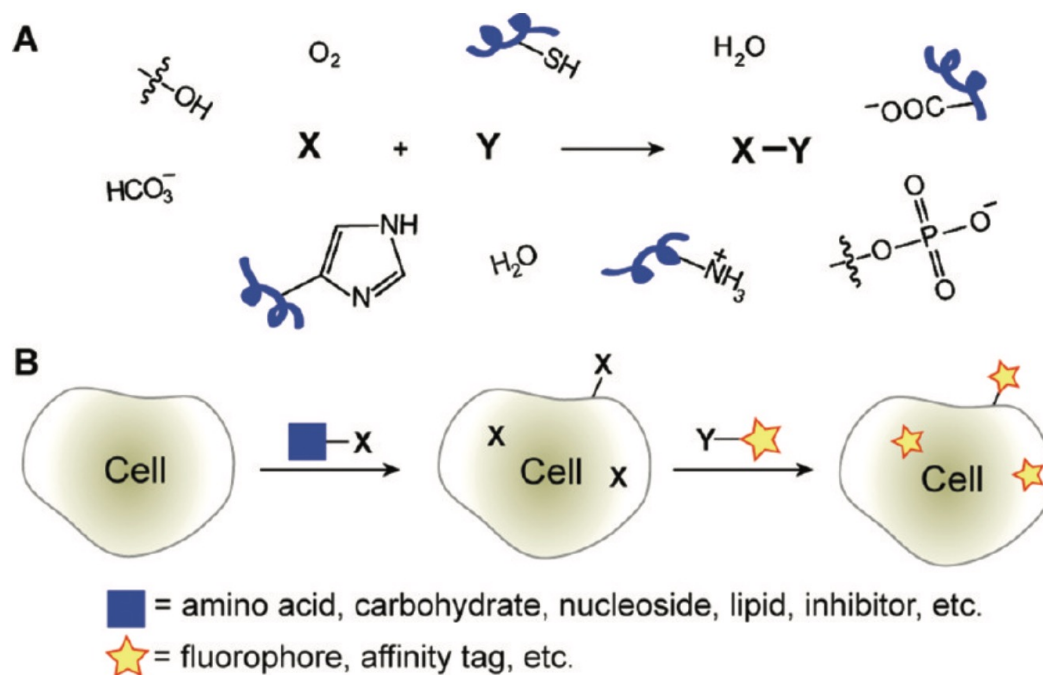


FIGURE 1. (A) A generic bioorthogonal chemical reaction between X and Y that proceeds in biological systems. (B) A common experimental platform for biomolecule probing using bioorthogonal chemistry. First, a non-native functional group, often called a “chemical reporter”, is installed in a biomolecule of interest. The modified biomolecule is subsequently labeled using a bioorthogonal chemical reaction.

All these functional groups
are present in our body

CHO

COOH

NH₂

SH

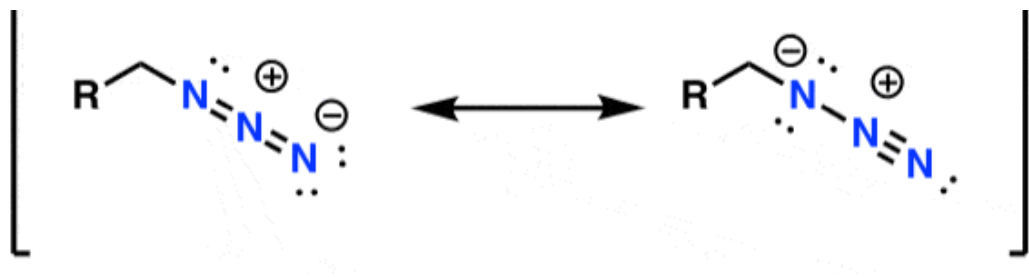
OH

CONH₂

COOR

C=C

There are many organic functional groups that are not present in our body



Azides: RN_3

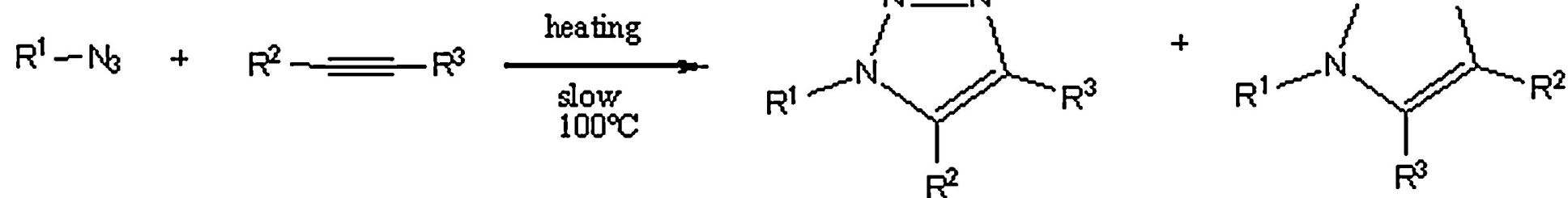


Alkynes: $\text{RC}\equiv\text{CH}$

A reaction classifies as “**bioorthogonal**” if it neither interacts nor interferes with a biological system

Huisgen 1,3-dipolar cycloaddition

Huisgen's 1,3-dipolar cycloaddition

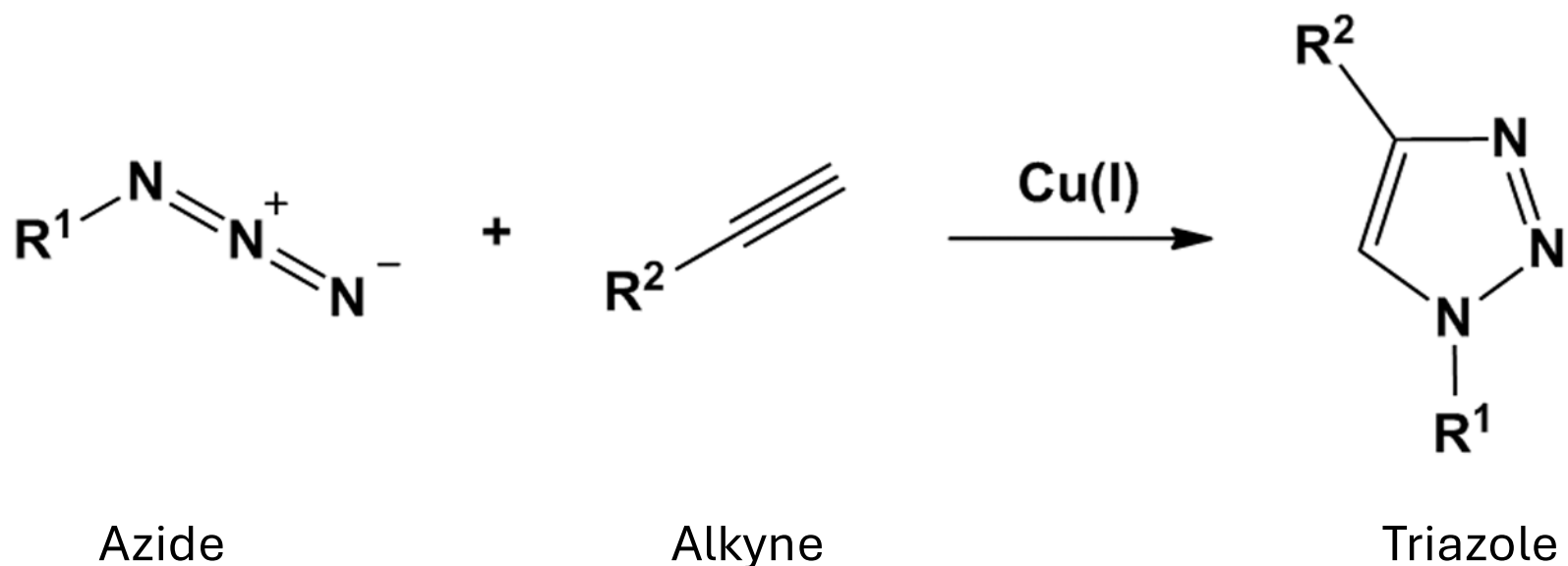


100 degree C

Both 1,4 and 1,5 isomers were formed

1961

Copper catalyzed Click Reactions



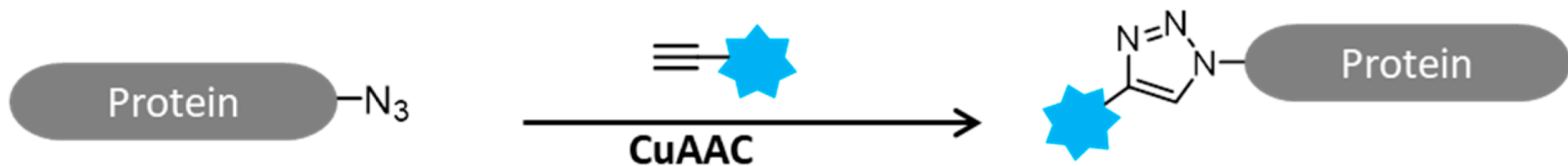
2001
2010

Copper alkyne azide cycloadditions (CuAAC)

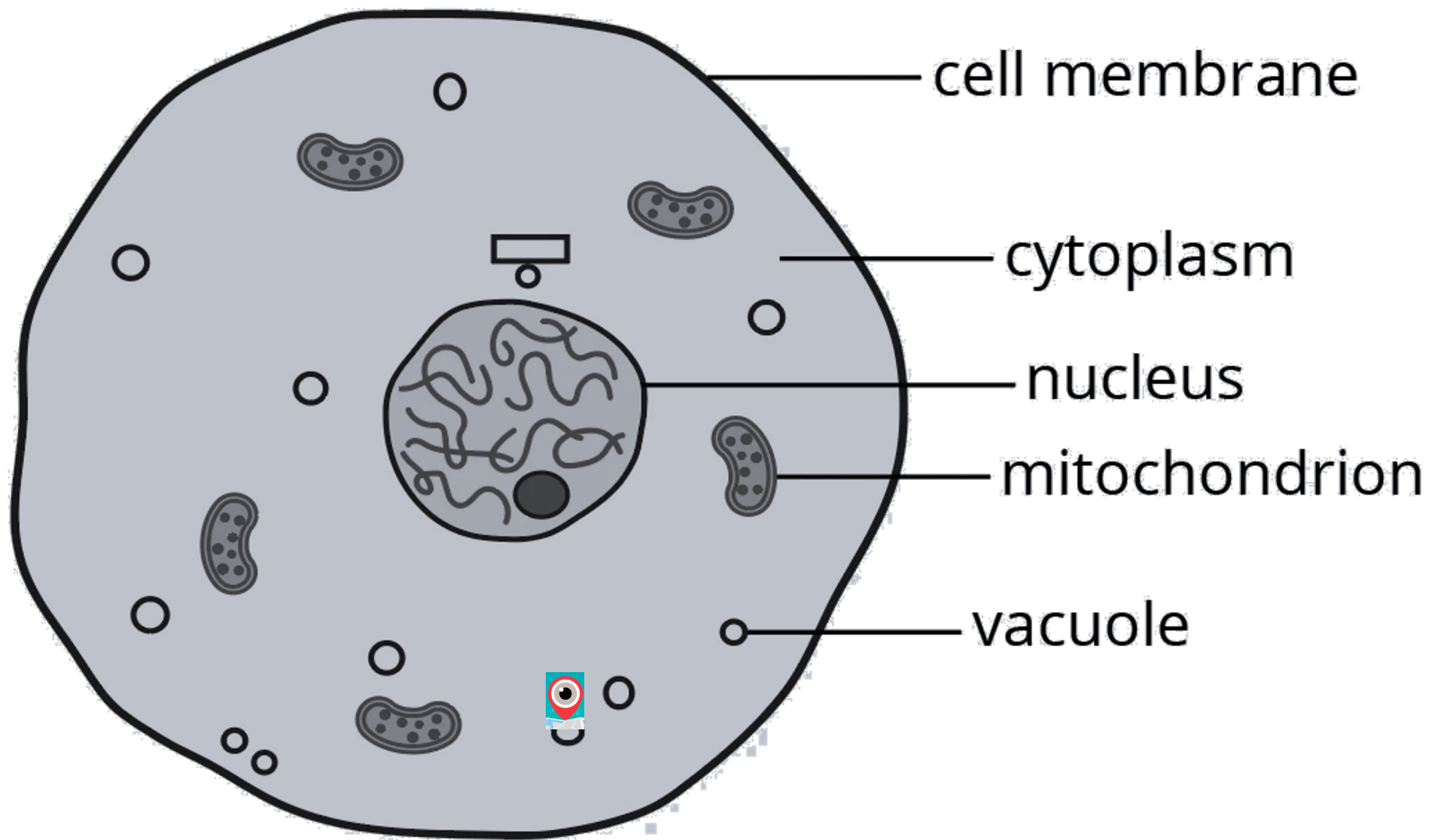
Room temperature
1,4 isomer only

Carolyn Bertozzi

Tag any biomolecules inside living cell using a fluorophore

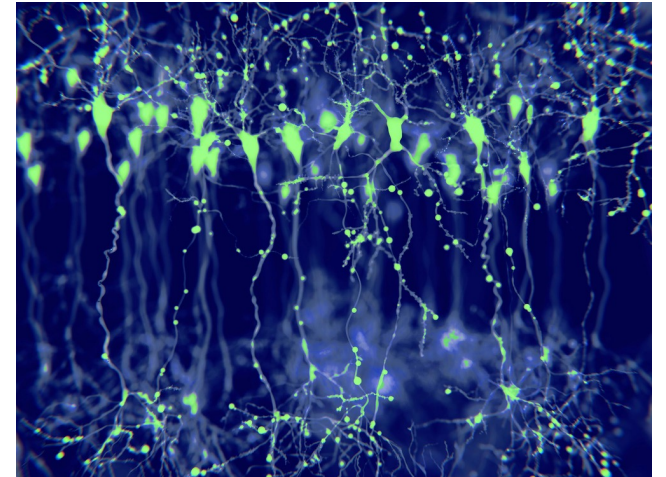
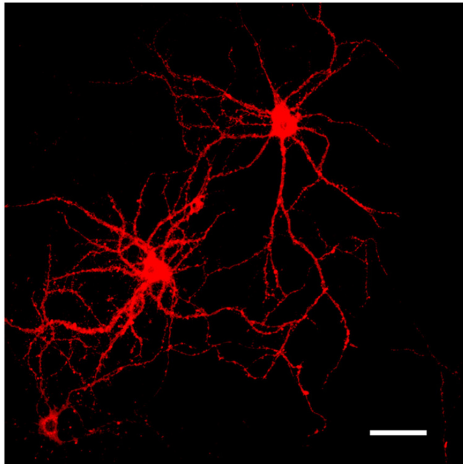
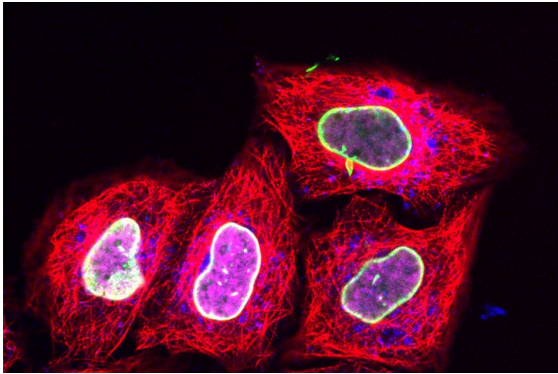


Sugar, lipids, proteins, peptides, DNA, nucleotides, metabolites



Live Cell Imaging – Confocal and fluorescence microscopy

Visualize cells and components using fluorophores



<https://www.youtube.com/watch?v=XDcf6P9Fpww>

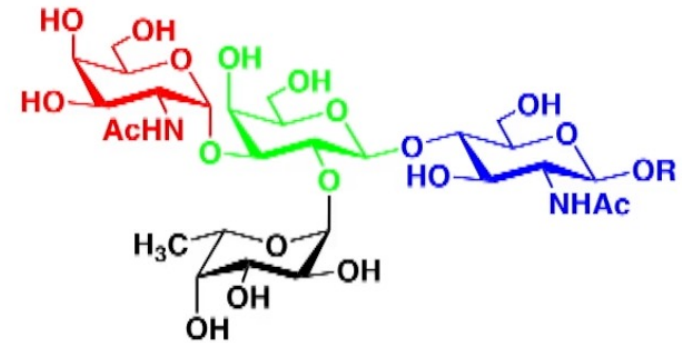
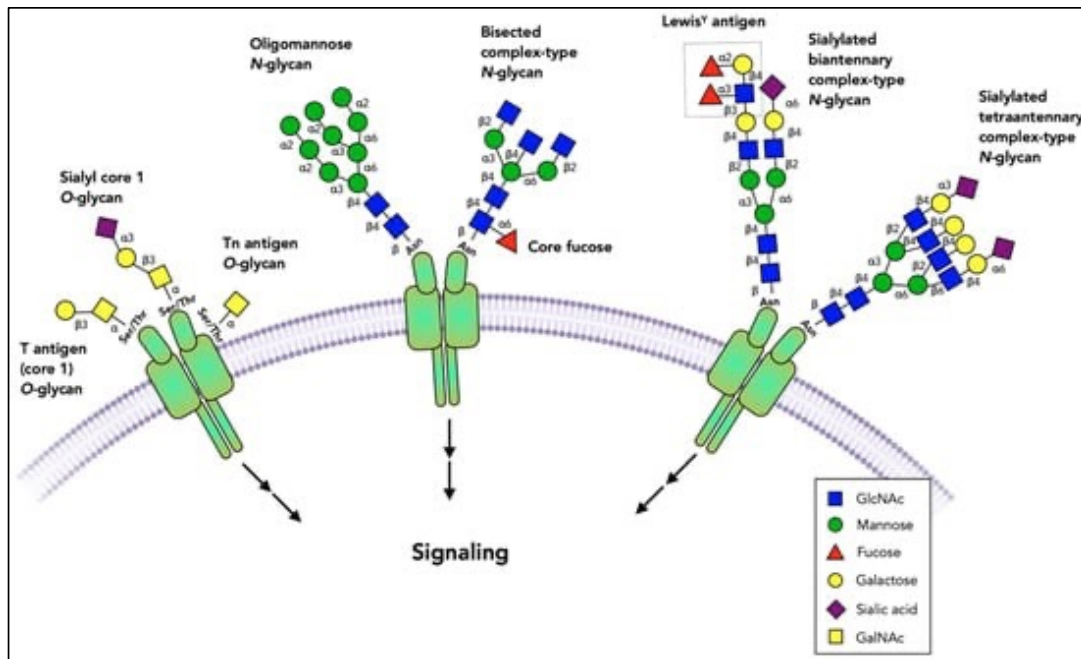
-
- Bioorthogonal Click Reactions

[Ted Talk by Carolyn Bertozzi](#)



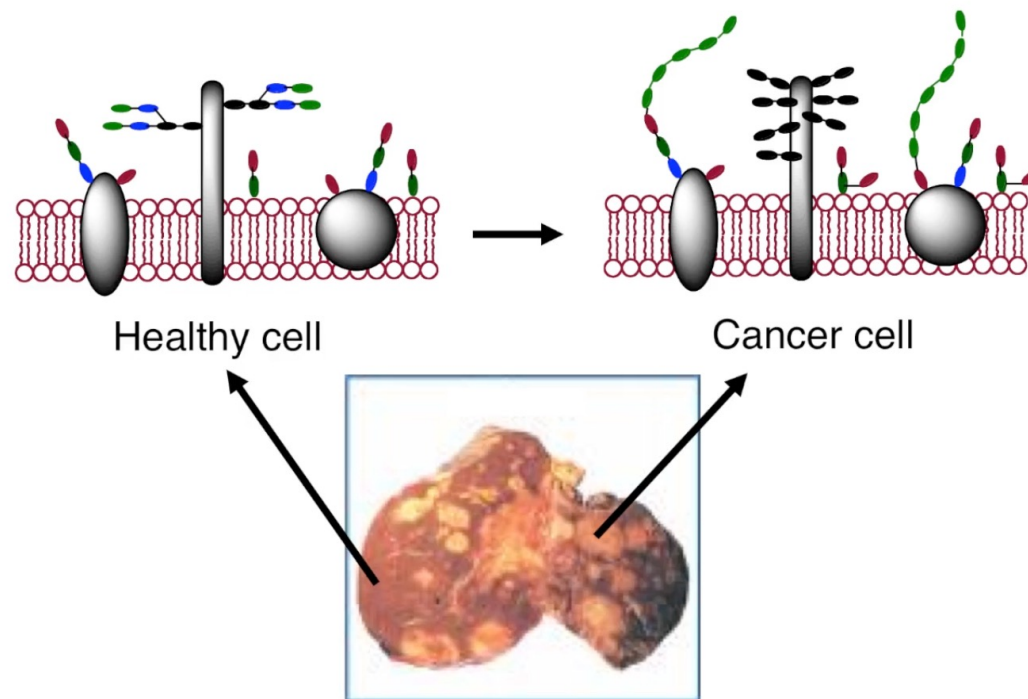
Stanford University, California, USA

Sugar molecules coat our cell surface



With the help of Bioorthogonal Click Reactions

Sugars change on cancer cells



Changing the sugar structure/ composition on cancer cells → cancer treatment